

Who Approves, Who Worries? A Dataset of Parent and Child Attitudes Toward Children's Use of Virtual Reality

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Highlights

Who Approves, Who Worries? A Dataset of Parent and Child Attitudes Toward Children's Use of Virtual Reality

- Introduces a novel dataset on both parents' and children's ethical perceptions of VR use
- Captures responses from 122 participants (67 children, 55 parents) across 52 families
- Combines Likert-scale ratings with semi-structured interviews across education, healthcare, and family scenarios
- Links demographic and socio-economic factors to acceptability, concerns, and perceived benefits of VR
- Provides an open, reusable resource with an example analysis and guidelines for secondary use

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ABSTRACT

As virtual reality (VR) becomes increasingly accessible to children in educational, healthcare, and family settings, understanding how different stakeholders perceive the ethical implications of VR adoption for children is critical. To support such investigations, this paper presents a dataset comprising responses from 122 participants (67 children aged 6–15 and 55 parents) who evaluated VR applications across three domains: education, healthcare, and family, using videos depicting both utopian and dystopian scenarios. The dataset includes demographic information, Likert-scale survey responses measuring acceptability, concerns, and perceived benefits, as well as semi-structured interview transcripts. As an example use case, we conducted an analysis examining relationships between selected demographic factors and perceptions of VR. We also discuss additional use cases and guidelines. This dataset provides a foundation for future research on individual differences in children's VR adoption and offers insights into how demographic and socio-economic factors shape ethical considerations surrounding children's use of VR.

1. Introduction

Virtual reality (VR) is becoming increasingly accessible, and research has shown the vast potential of VR in reshaping the way children play [1], learn [2, 3, 4], socialize [1], and receive health treatments [5]. Many promising VR applications for children have also emerged. For example, VR can enable fun, immersive, and interactive experiences beyond conventional entertainment, such as 2D video games or activities in the real world, which makes it highly attractive to children [6]. In educational settings, the 3D nature and interactive capability of VR can not only enhance children's engagement over traditional classroom formats [7, 8], but also visualize abstract concepts in a way that is easier for children to comprehend [9, 7]. VR has also been shown to be helpful in healthcare settings to improve both the effectiveness of the treatment and children's compliance [5], such as being used as a distraction for pain management [10, 11], motivation for exercises [5, 11], and as therapies for mental health issues [12] or neurodevelopmental disorders [11]. At the same time, VR deployments with children call for careful design and evaluation, as children are a protected population and may experience different risks, constraints, and benefits than adults [13]. The use of VR is known to be associated with risks and side-effects such as motion sickness [14, 15], eyestrain [16], and potential injury caused by collisions with real world obstacles [17], in addition to the more generic concerns about children's use of digital devices, such as addiction, privacy, safety, and content appropriateness [18, 19].

Prior studies primarily looked into either perceived or empirically validated benefits and risks associated with the use of VR for children [18, 19]. These studies identified important factors in shaping the general public's opinion towards VR for children. However, the acceptance and adoption of novel technologies ultimately depend on human perception and decisions, which are largely affected by one's cultural, educational, and socio-economic background [20, 21] and gender [22, 23], yet few studies have focused on users and specifically looked into how

demographic characteristics (e.g., age, gender) and socio-economic status (SES) (e.g., income, education) shape perceptions of using VR with children. While prior work has produced many datasets capturing ethical concerns around children's access to and use of mobile devices and digital media [24, 25, 26], there is still no comparable dataset focused on children's use of VR. Such datasets can be particularly valuable for investigations of open-ended questions in this emerging field, such as the impact of demographics, which can help provide additional insights into the adoption of VR for children and also reveal important implications for the generalizability of prior findings from smaller-scale studies with potentially limited demographic diversity.

To encourage and support a broader investigation of how demographic characteristics and SES relate to perceptions of VR use with children, we present an anonymized dataset consisting of the demographic characteristics and SES of 122 participants (parents and children). The dataset links participants' demographic/SES information with their reported acceptance, concerns, and perceived benefits of child VR across multiple use scenarios (education, family, and healthcare), captured through both Likert-scale survey responses and de-identified interview transcripts. We also demonstrate an example usage of the dataset by conducting a cumulative link mixed effect model to examine the impact of various factors previously observed to affect acceptance for novel technologies. We contribute to the child-computer interaction (CCI) community a reusable empirical resource that combines parent and child perspectives and supports cross-group comparisons, secondary analyses of how perceptions of VR use with children vary across individuals, and how these differences might relate to demographic characteristics and socio-economic status (SES), as well as follow-on studies on child-focused VR design and adoption.

2. Related Work

2.1. Virtual Reality for Children

Prior work in CCI has examined VR in child-focused contexts where immersion and 3D interaction can change how children engage with content, environments, and peers. For example, in educational contexts, studies report that VR can support the teaching of abstract subjects such as science and mathematics by making concepts spatial and interactive, helping children engage with ideas that are difficult to convey through flat media [27, 28]. In elementary classrooms, immersive virtual environments have also been used to increase motivation and participation through interactive activities integrated into instruction [29, 30]. Additionally, VR-based collaborative task designs have been shown to enhance elementary students' teamwork and creativity, providing more interactive experiential learning processes [31, 32]. VR can also create shared, immersive spaces that support parent-child communication and peer collaboration. Research indicates that integrating VR with physical tools (e.g., LEGO bricks) promotes communication and coordination during cooperative tasks by requiring children to negotiate shared spatial decisions and jointly manipulate both virtual and physical elements [33, 31, 34]. Such hybrid activities encourage them to express intentions,

seek assistance, and coordinate with parents or peers, thereby fostering richer coordinated patterns of interaction. Furthermore, social VR environments presented to children are recognized for enhancing teamwork abilities and promoting their social development [35, 3]. The immersion and engagement that VR affords also demonstrate significant value in medical settings for children. For instance, VR can effectively alleviate anxiety and pain during medical procedures, particularly during needle pricks and dental treatments [36, 37, 38, 10]. VR-based exercises are also preferred by obese children over conventional video games, potentially improving their adherence to regular exercise [5].

This body of work motivates the need to understand stakeholder perceptions of child VR as these applications move from research prototypes toward broader use. This paper contributes an anonymized dataset that captures parent and child perceptions across different child-VR scenarios and links them to demographic characteristics and socioeconomic status (SES), including both Likert-scale responses and de-identified interview transcripts.

2.2. Ethical Concerns in Children's Use of VR

Although VR has been explored for a range of child-focused applications, researchers have also highlighted ethical and safety concerns around children's VR use [36, 37]. Synthesizing evidence across studies, Kaimara et al. identify concerns spanning physical comfort and safety (e.g., cybersickness and other adverse bodily effects, eye-related discomfort, and collision or injury risks), behavioral and cognitive impacts (e.g., problematic use and potential difficulty for younger users in interpreting highly realistic virtual experiences), and broader ethical issues (e.g., privacy/data practices, content appropriateness, and informed use) [13]. These issues can be amplified in social VR, where embodied interaction and a strong sense of co-presence may increase the harm of harassment or other misconduct [39, 40, 41]. Prior work has also highlighted ethical issues related to equity and inclusion in children's access to VR technologies. Educational and developmental VR opportunities may be unevenly distributed due to socioeconomic status, and existing systems often provide inadequate support for children with disabilities [19]. Despite growing attention, child-specific guidance and safety practices remain limited [42].

Commercial VR ecosystems reflect these concerns through age thresholds and parent-facing controls, but policies vary across platforms. For example, Sony states that PlayStation VR2 [43] is not for children under 12, and PICO [44] states that users must be at least 13 to use its device (with higher minimum ages in some jurisdictions). Major social VR platforms also set age limits (e.g., VRChat [45] requires users to be 13+) . Meta's Quest [46] ecosystem allows preteens via parent-managed accounts for ages 10–12 and provides supervision tools (e.g., app controls and account-level settings) through its Family Center resources . However, platform rules and control features do not replace research-based guidance for younger children, and actionable safety guidelines that are specific to children remain limited [42].

Although prior work has examined ethics and safety concerns around children's VR use, much of this evidence is scenario-specific and often centers adult viewpoints (a few exceptions are [47, 48]), with many studies relying primarily on qualitative methods (e.g., semi-structured interviews [49, 50], focus groups, and observational studies [51, 52, 53]). To complement this line of research, we contribute a dataset that captures perceptions of child VR across three scenarios and includes both parents and children. The dataset combines Likert-scale measures of acceptance, concerns, and perceived benefits with de-identified interview transcripts, enabling both cross-participant quantitative analyses and deeper qualitative interpretation grounded in children's and parents' accounts.

2.3. Factors Influencing Technology Acceptance and Concerns

People's acceptance of new interactive technologies depends on how they interpret the technology's value, effort, and fit for their daily lives, and how they weigh potential risks. Prior work shows that these judgments are not uniform: they are shaped by cultural and educational context, SES, and other background factors that affect what technologies people encounter and how they evaluate them [20, 21]. Demographic characteristics further shape acceptance and concern patterns. Across studies of emerging technologies, gender and age have been linked to systematic differences in attitudes and risk perceptions [22, 54, 55, 23]. For example, Gonzalez et al. report that women and older adults show lower acceptance or interest in technologies such as AI and robotics [54]. Related work suggests women may be less optimistic and more risk-averse toward new technology products [56], and similar gender-linked patterns appear across other technology adoption contexts [57, 58, 59]. Age effects are also widely observed, with older adults less likely to adopt new technologies [60, 61]; proposed mechanisms include lower confidence and knowledge about new media [62, 63] and barriers to acquiring skills due to physiological and cognitive factors [64]. Importantly, demographic effects can interact: Morris et al. found that gender differences were more pronounced among older workers, whereas technology adoption among younger workers was closer to gender-neutral [65]. In child-related decisions, life stage and caregiving roles can also shape what adults view as "acceptable" use, because responsibility and supervision demands are part of the evaluation.

SES is often intertwined with practical constraints that directly affect adoption. Differences in income, education, and resource access can shape whether families can obtain devices, the time and support they can provide during use, and the alternatives available to them (e.g., other activities, childcare options, or school-provided resources) [20, 21, 66, 67]. In child-focused settings, these constraints can translate into different expectations about supervision, acceptable cost, and acceptable risk, since feasibility and safety are evaluated alongside potential benefits. Technology-related background also matters. Prior experience with related technologies can change what users expect, how quickly they learn interaction patterns, and what they interpret as tolerable side effects versus unacceptable harm. Prior experience has also been associated with acceptance of emerging technologies in multiple contexts [66, 67]. For child technology

use, perceptions are additionally shaped by family dynamics: children's views do not always align with parents' views [68], and prior work documents tensions where children are attracted to digital media while parents take on a mediation role [69, 70, 71]. At the same time, parents' attitudes and practices can influence children's attitudes and usage patterns [72, 73].

However, empirical resources that connect these factors to child-focused VR perceptions (especially across multiple scenarios and from both parent and child perspectives) remain limited. We respond to this gap and contribute a dataset that links demographic/SES information with stakeholders' perceptions of child VR. This dataset is intended as a reusable resource for the CCI community, enabling future researchers to apply their own statistical models and analytic assumptions to test hypotheses about factors in perceptions of child VR.

2.4. Datasets on Technology Usage for Children

Within the field of CCI and related research, several data resources [74, 75, 76, 77] have been published to document children's exposure to and use of digital technologies in daily life. These datasets typically focus on technology access pathways, usage patterns, and media consumption within family or social contexts, providing a macro-level empirical foundation for understanding children's digital lives. The EU Kids Online [74] project provided large-scale, cross-national survey data that systematically documents children's internet usage, online activities, and associated risks. It also collects feedback from both children and their parents, thereby supporting analysis of technology use and risk perception at the household level. The UK communications regulator Ofcom's report [75] based on national survey data describes children's and parents' usage and attitudes toward various digital media and technologies, providing a crucial reference for studying the social contexts and family differences in children's technology use.

The previous works primarily focus on technology use and media exposure. Most of them lack nuanced depictions of how children and their caregivers understand, evaluate, and weigh the potential impacts of digital technologies, particularly judgments at the cognitive and ethical levels. In contrast, the CCI area has also produced smaller-scale yet more nuanced datasets focused on cognitive and interactive dimensions. These datasets directly examine children's behaviors, perceptions, and social relationships during specific technological interactions. For instance, a dataset introduced in prior work [76] centers on children's interactions with mobile devices, incorporating age-related characteristics to provide reusable resources for studying differences in how children engage with computational devices. Additionally, the DAMI-P2C dataset [77] captures emotional expressions and collaborative relationships within parent-child dyadic interactions, supporting analyses of children's technology engagement within cognitive and social contexts from a dyadic perspective.

Existing datasets either focus on children's technology use and media exposure at a large-scale level or concentrate on behavioral and cognitive analyses within specific interaction contexts. In contrast, publicly available datasets

that systematically examine children and their parents' cognitive and ethical responses to emerging technologies, particularly immersive VR, across diverse life scenarios remain extremely limited. Because emerging technologies such as VR have not yet reached widespread adoption, large-scale data collection efforts often lack access to a sufficient pool of participants with meaningful hands-on experience. This challenge is further amplified for children, whose access to VR is even less reliable to assume in a large, self-selected sample. Additionally, VR's embodied interactions, high immersion, and potential safety and developmental risks present unique challenges in childhood contexts distinct from other digital media. To address these gaps, the research team constructed a dataset through live VR demos and in-person interviews that systematically documents parents' and children's acceptability, concerns, and perceived benefits regarding VR use across educational, healthcare, and family settings. These attitudes are combined with demographic characteristics and family role information to support future in-depth research on cognitive and ethical issues in children's VR usage.

3. Dataset of Attitudes Toward Children's Use of VR

The dataset was originally collected for a study examining parents' and children's ethical concerns about children's use of VR [19]. The prior publication reported a qualitative analysis of perceived risks and design considerations. Building on that analysis, this work releases the de-identified survey and interview dataset, documents its structure and conditions for secondary use, and presents an analysis of demographic and family-role associations as an example use case.

3.1. Methods for Data Collection

3.1.1. Participants

The dataset contains records from 122 participants recruited in 52 family groups, including 67 children and 55 parents¹ from 52 families. Four children did not complete the study so only their partially collected data were retained.

In terms of the demographics of the adult participants within the dataset, guardians (primarily parents) accounted for 74.54% (N = 41), while 7.27% were siblings (N = 4), 7.27% were other direct family members (N = 4), 5.45% were grandparents (N = 3), and 5.45% were other non-family members (N = 3). The average age of adult participants was 42.07 years (SD = 8.94), including 32 females and 23 males. Regarding the ethnicity of parents, 81.82% identified as White (N = 45), 7.27% as Latinx (N = 4), 5.45% as Black (N = 3), and 5.45% as other racial backgrounds (N = 3). About 36.36% of the adult participants (N = 20) had not completed a college degree, 18.18% (N = 10) reported a household income below \$75,000. Additionally, 41.82% (N = 23) had previous experience using VR technology.

¹While the term "parents" typically refers to mother and father, it is used in a broader sense in this work and includes other guardians and adult family members (e.g., grandparents, siblings)

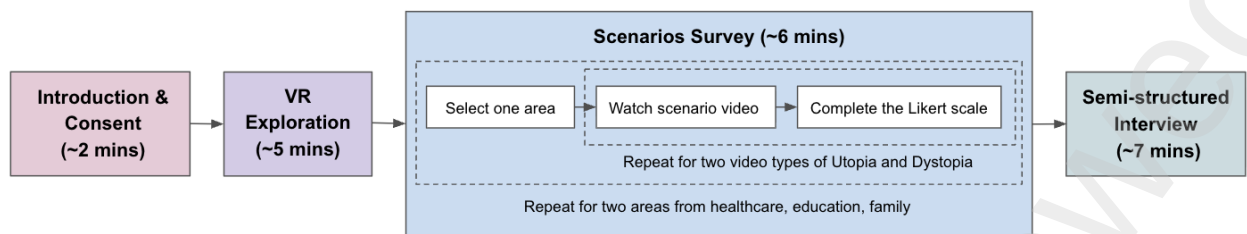


Figure 1: Overview of the study procedure

The children's average age was 10.46 years ($SD = 1.9$), including 38 females and 29 males. The target age range for participation was 7–13 years, but exceptions were made for children slightly outside this range. For instance, if a child was eligible and agreed to participate in the study, their siblings were allowed to participate as long as their ages were not too far off the target range. As a result, one 6-year-old and one 15-year-old child were included. Among the children, 76.12% identified as White ($N = 51$), 10.45% as Black ($N = 7$), 7.46% as Latinx ($N = 5$), and the remaining 5.97% ($N = 4$) were classified as other ethnicities. Notably, 44.78% of the children reported having prior experience with VR.

3.1.2. Study Design

Data collection for the original study took place at a public-facing research facility on the grounds of a large state fair in the U.S. Midwest. Recruitment was conducted on site. Research staff invited passing groups that included at least one child to participate, and interested groups were enrolled after eligibility was confirmed. Before evaluating the study scenarios and completing the surveys and interviews, participants took part in a brief guided VR familiarization activity to ensure that they had first-hand VR experience. The study was approved by the university's Institutional Review Board (IRB). All participants received a backpack or a stuffed toy as a token of appreciation for their involvement.

While online surveys and interviews allow for more efficient, larger-scale data collection, data were collected in person so that participants' responses could be grounded in first-hand experience with VR. Unlike conventional digital media such as smartphones, tablets, or computers, VR remains relatively unfamiliar to the general public, so online recruitment for a VR-related study risks over-representing existing VR users or enthusiasts relative to the broader population, who may have little to no exposure or interest in the technology. In-person data collection also allowed the researchers to give every participant hands-on experience with VR before they responded, which is critical to ensure that all responses reflected some direct exposure rather than pure imagination.

3.1.3. Artifacts

VR Exploration System A VR exploration system was developed to give participants, especially those who had no prior VR experience, an opportunity to try this technology. This system comprises five 360-degree video clips,

each showcasing different rooms and astronauts' activities within the space station. Each clip has a duration of approximately 1.5 minutes and was sourced from the European Space Agency's International Space Station tour ². In the VR environment, participants are represented by a neutral, robot-like avatar, separated by colors. They can utilize the trigger button on their controller to select the room they wish to enter, aligning with the door shown in the video content. Notably, using this button allows them to bring their partner(s) into the same room, ensuring that the entire group experiences the synchronized video context.

The system was pilot-tested with eight college students to ensure that the system is user-friendly and can be operated without the need for training. Based on their feedback, button placements and design were refined, and the controller interaction was simplified to use a single button for switching between rooms.

Scenarios Scenario development followed an anticipatory ethics approach commonly used to examine ethical questions associated with emerging technologies [78]. Six scenarios were developed to describe possible future uses of VR across three domains: education, healthcare, and family life. These domains were selected based on prominent applications of VR for children identified in prior research, as discussed in Section 2.1.

As technological changes are often conceived of as variables with at least two potential outcomes [79], two narrative scenarios for each area were developed, showcasing both potential positive impact (utopian) and negative impact (dystopian). These scenarios act as intuitive stimuli and provide a foundation for thinking about and discussing the ethical issues encountered during the study. It is important to note that the scenarios were not explicitly labeled as utopian or dystopian, allowing participants to form their own interpretations and judgments. A summary of these scenarios is listed below. Full scenarios can be found in Figures 3 to 8 in appendix A.

- Education Utopia (EU): VR experiences augmenting an after-school STEM program.
- Education Dystopia (ED): VR shifts to education entirely within a virtual school setting.
- Healthcare Utopia (HU): Use of VR for distraction during children's medical MRI procedures.
- Healthcare Dystopia (HD): Children using VR to cope with pain and avoid physical exertion in post-medical procedures.
- Family Utopia (FU): Family members engaging together in VR gameplay.
- Family Dystopia (FD): Family members individually immersed in separate VR experiences while together physically.

²https://www.esa.int/ESA_Multimedia/Sets/Space_Station_360_ISS_tour



Figure 2: A parent and a child trying the VR Exploration System

3.1.4. Study Procedure

To collect this dataset, each session lasted approximately 20 minutes in total (See Figure 1). The researcher first introduced the study to the guardians and asked them to provide informed consent for both themselves and their children. Guardians were also asked to fill out demographic questionnaires on a tablet. Meanwhile, another researcher explained the consent form orally to the children to obtain their verbal assent and answer any questions they might have.

To familiarize participants with VR, researchers guided the children and their guardians to explore a 360-degree VR environment using the VR exploration system described in section 3.1.3. They were assisted in adjusting their headsets and controllers. During the exploration, participants were encouraged to describe their observations and ask each other questions. This step took approximately five minutes. See Figure 2 for a visual illustration.

Next, participants were asked to select one of three areas of interest, i.e., education, healthcare, or family. Participants then viewed one video depicting the utopia scenario and one depicting the dystopia scenario within the chosen domain (in randomized order), each lasting about one minute. After watching each video, participants answered three Likert-type questions designed to assess their attitudes toward the use case presented in the video (see section 3.1.5). Then the same procedure was repeated for each participant regarding their second choice of area of interest. This took about six minutes.

Finally, participants underwent separate but parallel semi-structured interviews (see section 3.1.5) for approximately seven minutes, with visibility to one another. These interviews were audio-recorded for subsequent analysis.

3.1.5. Measures

Survey For each scenario associated with the chosen areas, participants provided Likert-type responses to three items corresponding to their acceptance, concern, and perceived benefits of VR, respectively. The adult- and child-facing items addressed the same three dimensions but used developmentally appropriate wording and response formats. The child-facing versions used simpler language to support comprehension while retaining the intended focus of each item.

The items for parents are as follows:

- I would approve of the use of VR as shown in this story by my child (*Q1*)
- I am concerned about the potential negative effects of VR as shown in this story (*Q2*)
- I believe that using VR as shown in this story would bring positive outcomes to children and families (*Q3*)

The questions for children are as follows:

- Would you want to use VR as shown in this story? (*Q1*)
- Do you think using VR as shown in this story might create problems in children's lives? (*Q2*)
- Do you think using VR as shown in this story would help kids and the world in positive ways? (*Q3*)

The Likert-type response format for adults involved indicates whether they strongly agree, agree, neutral, disagree, or strongly disagree with each item. Items were randomized within the survey. To design the Likert scale for children, researchers adhered to the recommended approach outlined in Royeen's work [80] and adapted the adult version to a three-option response format: *No*, *A little*, and *A lot*. The three options were represented by progressively larger blocks measuring 2×2 , 2×3 , and 2×4 inches. The visual size cues were intended to help children distinguish among the response options, including those who experienced difficulty reading the written labels.

The item wording and response presentation were refined through an iterative pilot study involving six adults and three children recruited through personal contacts. During the first stage, three adults completed the full study procedure and provided feedback on the instructions, item wording, and response format. Revisions were made based on this feedback. During the second stage, the revised materials were evaluated with the remaining three adults and three children. Feedback from the child participants informed further revisions to the child-facing wording and response presentation. The final versions were selected based on participants' reported interpretations of the items and their ability to distinguish among the response options.

Semi-structured interview Interviews were conducted separately and individually for each participant. During the interview, the researcher initiated the conversation by reminding the areas participants previously chose. Participants were asked about their thoughts on the use of VR in the stories presented to them and what aspects of these stories stood out. For example, participants were asked about benefits, potential issues or problems that VR might create for their children and families. Participants were also asked who should be responsible for deciding which VR experiences are suitable for children, and the considerations that creators should keep in mind when designing VR for children and families. Finally, interviews were conducted by inviting participants to share any additional insights or thoughts about VR for families.

Children were asked similar questions but conveyed using child-friendly wording, and the readability and clarity were tested with the same procedure as the same pilot study for the survey questions.

3.2. Overview of the Dataset

The dataset consists of 2 grouping variables and 27 attributes divided into 3 categories, i.e., sociodemographic attributes, geographic attributes, and responses. Additionally, interview transcripts are provided as separate files for each participant. Direct identifiers were removed from the survey records and interview transcripts. The research team also reviewed the transcripts for potentially identifying contextual information, including participant names, specific geographic locations, and uncommon family details, and removed or generalized this information when necessary. Exact ZIP codes are not released; the dataset includes only the derived Rural–Urban Continuum Codes and NIH rural–urban classifications. Child racial and ethnic identity is also excluded as a participant-level variable because the small sizes of several groups could increase re-identification risk while providing limited support for subgroup analysis. The de-identified dataset is publicly available on Github³. Original audio recordings are not included in the released dataset.

Grouping Attributes The grouping variables include **ID**, which is the unique identifier for each participant, as well as **Family-ID**, where participants sharing the same **Family-ID** belong to the same family.

Sociodemographic Attributes The sociodemographic attributes encompass both demographic characteristics and SES. Specifically, they include **Family-role** (parent or child), **Age**, **Gender**, prior **VR-experience** (yes or no), **Education** (highest level of education received), and **Income** (annual household income). Note that **Education**, and **Income** are only available for adult participants (parents) as they are not directly applicable to children.

Geographic Attributes In addition to sociodemographic attributes, the dataset also includes the rural-urban context for the families by indexing the Rural–Urban Continuum Codes (**RUCC**) and NIH’s rural–urban classification for

³<https://anonymous.4open.science/r/VR4Children-Ethics-Dataset-77C6>

health research (**NIH**) based on the zip-codes participants provided for their home. While both **RUCC** and **NIH** capture the degree of urbanization in an area and may be correlated, the former is focused on adjacency to metropolitan areas, whereas the latter primarily considers health-related service access. Both indicators were therefore retained to support research questions with different analytic goals.

Responses Responses contain participants' Likert-type responses to the 6 scenarios (EU, ED, FU, FD, HU, HD) $\times$ 3 items (acceptance, concern, and perceived benefits). The associated attributes are denoted as **EU-Acceptance**, **ED-Acceptance**, **ED-Benefits**, etc. Parents' responses were in 5-point-Likert scale. For the children, following the method recommended by Royeen [80], researchers adapted the scale to a three-option format: no, a little, a lot. Note that for each participant, only the responses associated with the two selected areas are available.

An additional attribute **Interview-Data** provides a reference to the interview transcripts in separate csv files for each participant. 12 participants' interview transcripts are unavailable due to recording issues.

4. Example Use Case of the Dataset

While the dataset supports a broad range of exploratory analyses, in this paper we provide one example use case by examining previously observed relationship between demographic variables and ethical concerns in the use of novel technology in prior studies.

4.1. Research Questions and Hypotheses

In this user case, we aim to investigate: (RQ1) how age and gender relate to perceptions of VR use with children, and (RQ2) how parents' perceptions differ from children's.

To answer these questions with the survey measures in the dataset, we translate each RQ into testable hypotheses about three outcome dimensions—*acceptability*, *concerns*, and *perceived benefits*. We formulate the following hypotheses:

- **H1:** Younger participants will have higher acceptability (H1.A), fewer concerns (H1.B), and more perceived benefits (H1.C) towards the use of VR than older participants.
- **H2:** Male participants will have higher acceptability (H2.A), fewer concerns (H2.B), and more perceived benefits (H2.C) towards the use of VR than female participants.
- **H3:** Children will have higher acceptability (H3.A), fewer concerns (H3.B), and more perceived benefits (H3.C) towards the use of VR than parents.
- **H4:** The “Utopia” video type will be associated with higher acceptability (H4.A), fewer concerns (H4.B), and more perceived benefits (H4.C) than the “Dystopic” video type.

H1 through **H3** were based on findings from prior literature [22, 54, 55, 23]. **H4** serves as a manipulation check for the two video framings (“Utopia” vs. “Dystopia”) to confirm that the stimulus types elicit systematically different responses.

For exploratory purposes, we also consider the differences in participants’ perceptions across the 3 areas, i.e., family, education, and healthcare, to examine if acceptability and concerns differ between applications.

4.2. Data Analysis

Cumulative Link Mixed Models (CLMM) were used to model the relationship between selected demographic variables and the Likert-type survey scores. CLMMs account for the ordinal nature of the Likert-type outcome variables, and captures the differences between individuals and families through random effects.

Since the scales for the Likert-type questions were different between the parents (5-point) and the children (3-point), we remapped the parents’ responses to a 3-point scale, where “*disagree*” and “*strongly disagree*” were mapped to “*no*”, and “*agree*” and “*strongly agree*” were mapped to “*a lot*”. This allows us to include parents and children in a unified model.

Three CLMMs were fitted using the Laplace approximation in R (version 4.3.1), each corresponding to one of the three survey questions. In the regression models, the fixed effects include demographic and study-specific variables previously associated with VR acceptance, such as *age*, *gender*, and *family role*, along with *scenario* and *video type*. To account for the correlation between age and family role (parents are older than children), we also included their interaction term ($age \times family\ role$). Random intercepts for *participant IDs* nested within *family IDs* were included, to account for the hierarchical structure of the data.

We conducted post-hoc pairwise comparisons with Tukey adjustment for *scenario*, which is categorical and contains more than 2 levels, because we wanted to explore the differences between all levels, rather than only contrasting against a reference level.

The *age* variable was mean-centered and scaled by standard deviation to reduce multicollinearity, and variance inflation factors (VIF) were computed to verify that no variable induced significant multicollinearity concern. The maximum standardized VIF was 2.8 for *age* and *family role*. The VIFs of all other variables were under 2. The number of observations for each *scenario* was as follows: Education ($n = 603$), Family ($n = 528$), and Health ($n = 306$).

4.3. Results

Convergence was achieved for all models, with maximum gradients well below standard thresholds ($< 10^{-3}$) and condition numbers indicating stable, non-singular fits (maximum $\text{cond.H} \approx 1200$).

4.3.1. Acceptability of VR

According to the regression result, male participants had 1.65 times the odds of reporting higher acceptability than female participants ($estimate = 0.50$, $SE = 0.25$, $Z = 2.03$, $p = 0.04$), supporting H2.A. The “utopia” condition had 4.22 times higher odds of receiving higher acceptance scores than the “dystopia” condition ($estimate = 1.44$, $SE = 0.21$, $Z = 7.01$, $p < 0.001$), supporting H4.A. No significant effect of age or interaction between age and family role was found, which does not support H1.A. Finally, parents had 27.94 times higher odds of reporting higher acceptability scores than children ($estimate = 3.33$, $SE = 1.25$, $Z = 2.67$, $p < 0.01$), contradicting H3.A.

Post-hoc analysis further revealed that participants had 2.53 times higher odds of reporting higher acceptance of VR use in the health scenario than in the family scenario ($estimate = 0.93$, $SE = 0.30$, $Z = 3.14$, $p < 0.01$), as well as 1.80 times higher odds than in the education scenario ($estimate = 0.59$, $SE = 0.29$, $Z = 2.06$, $p = 0.098$), although the latter did not reach the significant threshold of 0.05.

4.3.2. Concerns about VR

The regression result indicated that participants had 5.81 times higher odds of reporting more concern about the “dystopia” condition than the “utopia” condition ($estimate = 1.76$, $SE = 0.20$, $Z = 8.78$, $p < 0.001$), supporting H4.B. However, no significant effects were found for age, gender, family role, or the interaction between age and family role, which does not support H1.B, H2.B, and H3.B.

Post-hoc analysis indicated 1.43 times higher odds of more concerns about the family scenario than the education scenario ($estimate = 0.36$, $SE = 0.21$, $Z = 1.72$, $p = 0.09$), but the difference was not statistically significant.

4.3.3. Perceived Benefits of VR

The analysis showed male participants had 1.80 times higher odds to be more positive about the benefits of VR compared to female participants ($estimate = 0.59$, $SE = 0.28$, $Z = 2.09$, $p = 0.04$), supporting H2.C. The “utopia” condition also had 4.39 times higher odds of being perceived as more positive than the “dystopia” condition ($estimate = 1.48$, $SE = 0.21$, $Z = 7.08$, $p < 0.001$), supporting H4.C. No significant effects were found for age, family role or the interaction between them, which does not support H1.C and H3.C.

Post-hoc analysis indicated 2.27 times higher odds of more perceived benefits of VR in the education scenario than in the family scenario ($estimate = 0.82$, $SE = 0.23$, $Z = 3.64$, $p < 0.001$), as well as 2.94 times higher odds of perceived benefits in the health scenario than in the family scenario ($estimate = 1.08$, $SE = 0.31$, $Z = 3.52$, $p < 0.01$).

4.4. Interpretation of Results

Male participants were found to be more acceptable towards the use of VR and perceived more benefits from it compared to female participants. This result is consistent with prior findings that males tend to be more positive toward

technology than females [22, 54, 55]. However, despite being more acceptable and perceiving more benefits, we did not find evidence that gender, or any of the other demographic variables considered, significantly impacted participants' concerns, which may indicate that the positivity does not necessarily come with less awareness of potential risks.

We did not observe the same trend reported in prior work that age negatively associated with acceptance towards new technologies [60, 61], since our statistical analysis failed to identify a significant effect of age on any of the outcome measures. A possible reason is that, due to targeting children within a specific age range, their parents were also within a relatively confined age range. Thus, despite age variances in the samples, there was not much generational difference within children or parents as in the participant pools in prior studies [60, 61], which led to the effect of age, potentially the effect of generation, not manifesting.

Contrary to our hypothesis that children would be more acceptable towards VR than their parents, our results indicated that parents showed higher acceptability than their children. One possible explanation is that parents may have a better understanding of and also the confidence to manage the use of VR to mitigate potential drawbacks, whereas children rely on their limited VR experience, and the scenarios presented were not the most attractive to them. However, because age and family role were structurally associated and had limited overlap between groups, their individual contributions could not be completely disentangled. Therefore, the estimated effects should be interpreted cautiously and should not be considered evidence of a purely parent-child difference independent of age.

Participants were consistently more positive towards the utopia condition than the dystopia condition, showing higher acceptability and perceived benefits, and fewer concerns about the use of VR. This confirms that the two videos effectively elicited the intended contrasting reactions by design.

Among the three scenarios considered in the study, participants appeared most cautious about the use of VR in the family context, reporting lower acceptability and perceived benefits, as well as greater concerns. One possible reason is that participants were recruited as family units, which may have led them to evaluate the family scenario more critically than the others. A possible explanation is that, similar to other digital technologies, the use of VR, if not properly managed, may detach users from the real world and hinder the maintenance of healthy interpersonal relationships among family members. However, because only two video examples were provided for each condition, the findings may not generalize to all potential use cases within these application scenarios. Future studies could further examine the differences in acceptability and ethical concerns surrounding the use of VR across these contexts.

5. Discussion

5.1. Opportunities for Future Use of the Dataset

In the previous section, we presented an example analysis to replicate prior findings of the influence of gender and age on acceptance and concerns toward novel technologies, specifically in the context of VR for children. Depending

on reference work, researchers can select different subsets of variables or adopt different statistical models, which may lead to different conclusions. This type of research allows comparison between the demographic influences on emerging technologies in general and the specific instance of VR for children, potentially providing valuable insights to understand the unique aspects of this specific technology applied in a specific setting.

Besides replication studies, the dataset also supports in-depth exploratory analysis that focuses on characterizing the differences in perception towards children's use of VR between various demographic groups without theory- or literature-driven hypotheses. These studies can be particularly useful for suggesting new trends that were previously unreported, although further confirmation studies are required to validate the generalizability of the findings.

Additionally, most empirical studies, similar to the example given in the previous section, report a single statistical model depending on the hypotheses and the goal of the study. However, each model only represents one of the many possible ways to describe the observed data, and there may exist alternative models that can achieve comparable or even better accuracy. The dataset can be used to test, compare, and evaluate different statistical models to uncover alternative interpretations of the observations.

5.2. Guidelines for Dataset Use

This dataset is intended to support secondary analyses of how stakeholders perceive VR use with children across scenarios. Below we summarize modeling considerations that are particularly important for valid inference and for making results comparable across future studies.

Check correlated predictors before fitting multivariate models. Several attributes are expected to be strongly correlated. For example, **RUCC** and **NIH** both capture aspects of geographic context, and **age** is naturally related to **family-role** (child vs. parent). Socio-economic variables such as **income** and **education** may also be collinear. We therefore recommend examining pairwise associations and/or variance inflation factors before fitting multivariate models, and reporting how correlated predictors were handled (e.g., dropping one variable, fitting separate models by construct, or using regularization).

Model parents and children with care (attribute availability and scale differences). The dataset includes both parent and child responses, but combining them within a single statistical model requires additional decisions because some parent attributes are not available for children. In particular, **income** and **education** are only collected for adults. If a unified model is desired, one reasonable strategy is to assign children the household **income** reported by their parent/guardian, since it reflects family-level resources. In contrast, **education** is not directly defined for children; researchers will need to either omit **education** from child-inclusive models, treat it as adult-only, or explicitly justify any placeholder strategy. A second issue is measurement comparability: parents responded on a 5-point Likert scale,

whereas children used a 3-option response scale. As noted in Section 4.2, pooling responses requires remapping the parent scale to a 3-point scale (or alternatively analyzing parents and children separately and comparing patterns at the level of estimated effects). When remapping, we recommend stating the mapping rule and acknowledging the trade-off that collapsing categories reduces resolution in adult responses.

Report analytic choices to support reuse and comparison. Because this dataset is meant to be reused, the interpretability of follow-on analyses depends on transparent reporting. We recommend that papers clearly document (1) whether parents and children were modeled jointly or separately, (2) how **income** and **education** were treated in child-inclusive analyses, and (3) how the response scales were harmonized (if applicable), alongside standard model diagnostics and uncertainty estimates.

5.3. Limitations of the Dataset

The scale of the dataset is relatively small compared to some of the related work discussed in section 2.4 [74, 75]. As discussed in section 3.1.2, emerging technologies such as VR differ significantly from established technologies in terms of familiarity and adoption. Online recruitment for a VR-focused study may disproportionately attract individuals who already use or are interested in VR, while responses from participants without direct VR experience may be based largely on conjecture. To reduce these risks, the research team recruited participants and conducted the study in person, ensuring that all participants experienced VR before completing the surveys and interviews. This decision increased the time and resources required for recruitment and consequently constrained the sample size. The smaller sample limits statistical precision and the power of subgroup and scenario-specific analyses.

Although the dataset includes variation across several demographic and family characteristics, participants were recruited in person at a large public state fair in the U.S. Midwest, and most were North American residents drawn from that region. This event-based recruitment strategy may also overrepresent families who attend public fairs and are willing to participate in a VR study. Consequently, findings from the dataset may not generalize to other regions of the United States or to other countries, where cultural norms, access to VR devices and services, regional economic conditions, and attitudes toward emerging technologies may differ. Analyses and interpretations should therefore account for the geographic and recruitment contexts of the dataset.

Another limitation is that the study design was relatively simple and less tightly controlled. Because the study sought comparable responses from parents and children, task complexity and the depth of the scenarios and questions were reduced to support comprehension among younger participants. This design improved accessibility and comparability across age groups but limited the depth with which participants' reasoning, attitudes, and responses to each scenario could be examined. For example, participants were allowed to select the two domains that interested them most. This choice was intended to elicit more meaningful responses than random assignment because participants,

particularly children, might have had limited knowledge of or interest in certain domains. However, the flexibility also led to imbalance across the three scenarios. Education was chosen by 101 participants, family by 88, and healthcare by 51. This results in unequal statistical power across different scenarios. Future studies can consider different study designs with different prioritizations while keeping this trade-off in mind.

Lastly, the qualitative part of the dataset is relatively limited. While the interview responses enable content analysis, some responses, particularly from children, lack the details and depth necessary for thematic analysis. Future work can consider improving study or question designs to motivate and elicit richer responses from children to support more in-depth qualitative investigations.

6. Conclusions

In this work, we presented a publicly available dataset comprising responses from 122 participants (67 children and 55 parents) who evaluated VR applications across education, healthcare, and family domains. The dataset combines demographic information, Likert-scale survey responses, and interview transcripts, providing a reusable resource that captures both parent and child perspectives on children's VR use. The example analysis illustrates how demographic and family-role variables can be examined using the dataset while also showing the importance of accounting for correlated predictors, group-specific response scales, and uneven domain selection. Despite limitations in sample size and geographic diversity, this dataset offers a foundation for future investigations into how demographic factors and family roles shape ethical considerations surrounding children's adoption of VR, supporting replication studies, exploratory analyses, and follow-up studies on child-focused VR design and adoption.

7. Selection and Participation

The study received Institutional Review Board (IRB) approval and was conducted at a public-facing research facility located on the grounds of a large state fair in the U.S. Midwest. Research staff recruited families on site by inviting eligible visitors who were accompanied by children. In total, 67 children and 55 adults from 52 family groups participated. Adult participants provided informed consent for their own participation, and a parent or legal guardian provided permission for each child's participation. Children provided verbal assent before taking part in the study.

CRedit authorship contribution statement

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Appendices

A. Scenarios

In this section we present the visual illustrations of the scenarios.

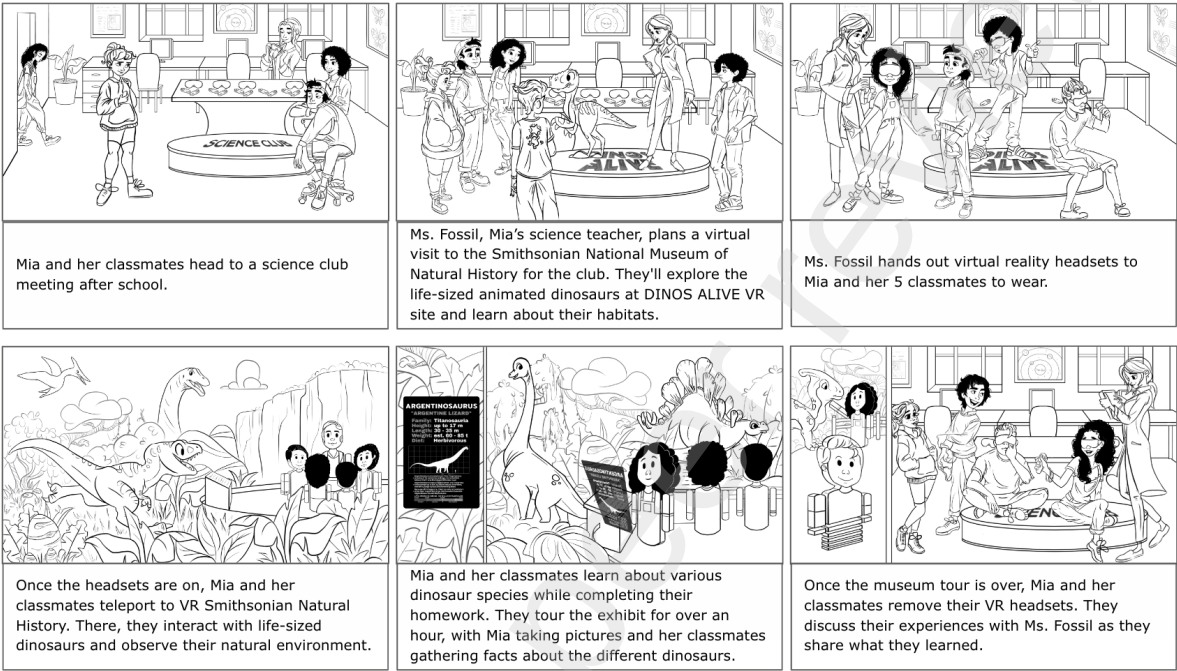


Figure 3: Education Utopia (EU): VR experiences augmenting an after-school STEM program.

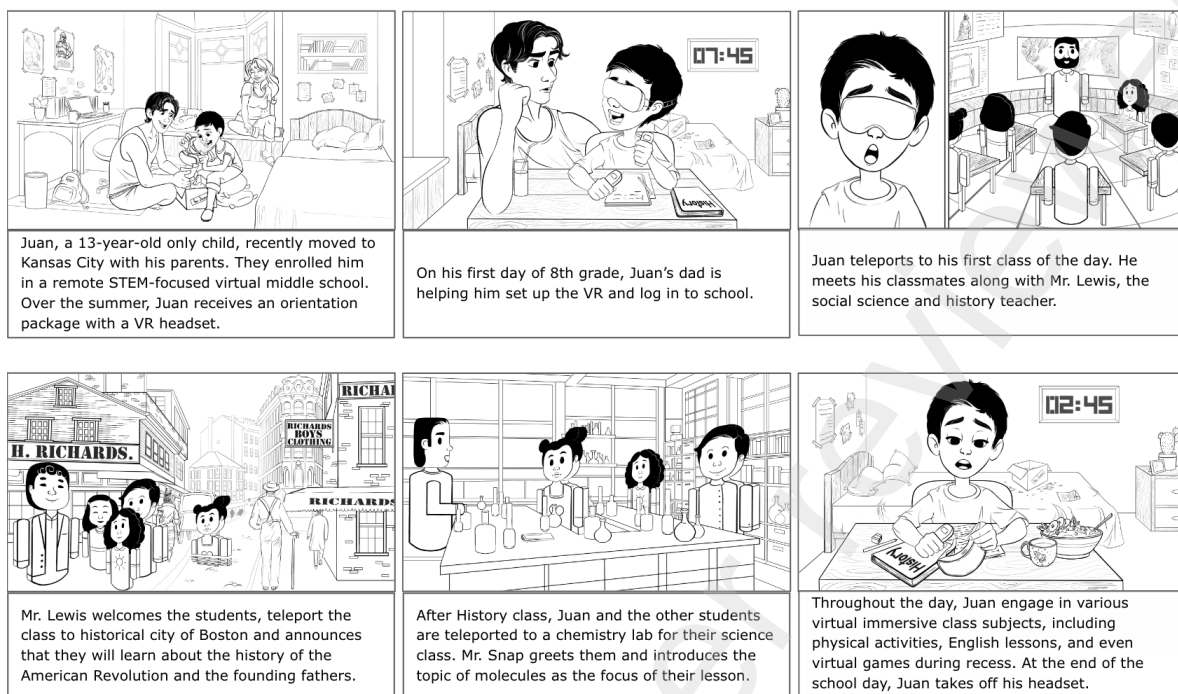


Figure 4: Education Dystopia (ED): VR shifts to education entirely within a virtual school setting.

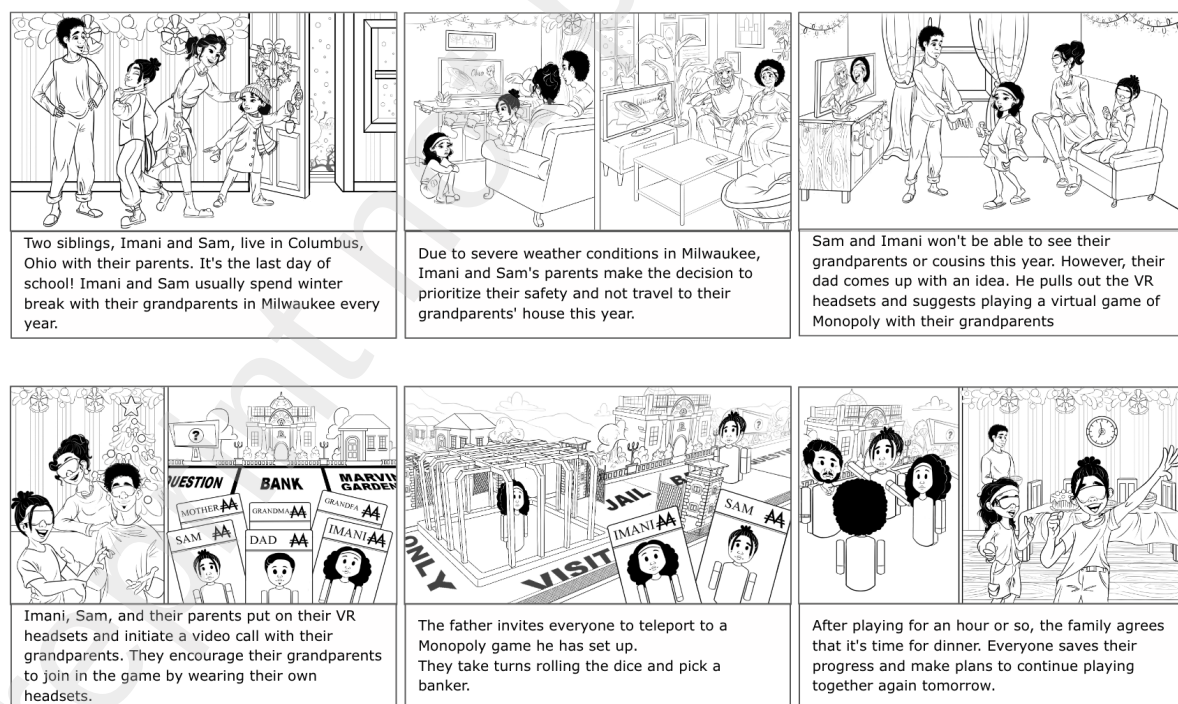


Figure 5: Family Utopia (FU): Family engaging together in VR gameplay.

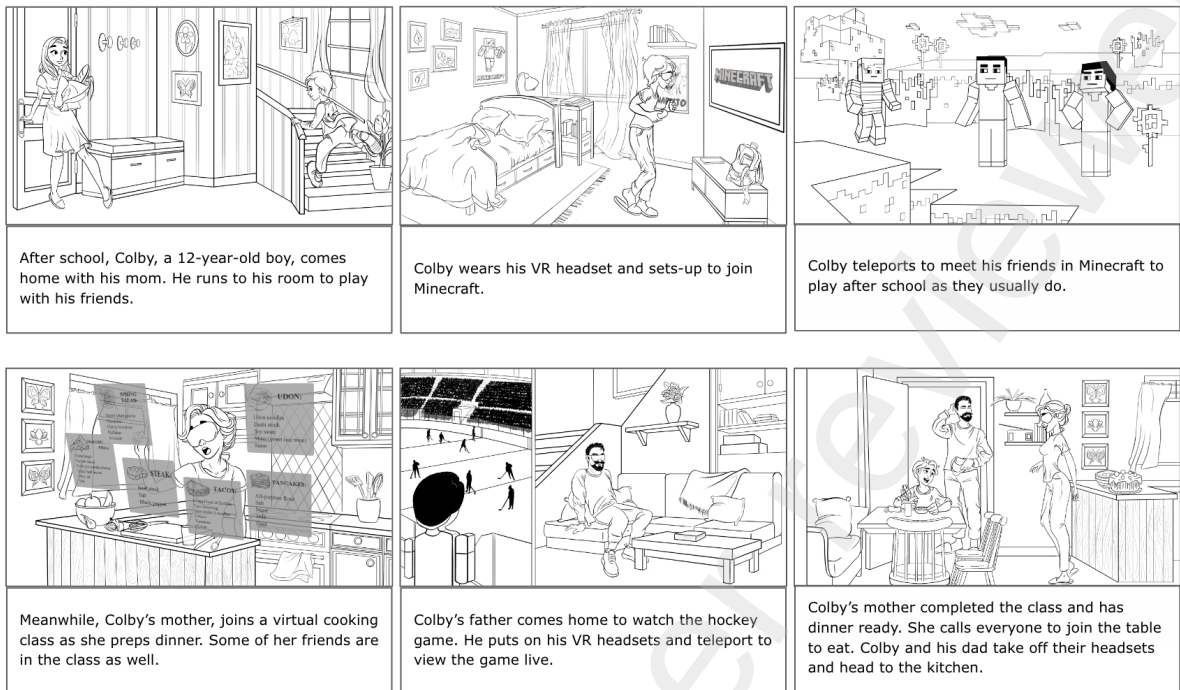


Figure 6: Family Dystopia (FD): Family members individually immersed in separate VR experiences while together physically.

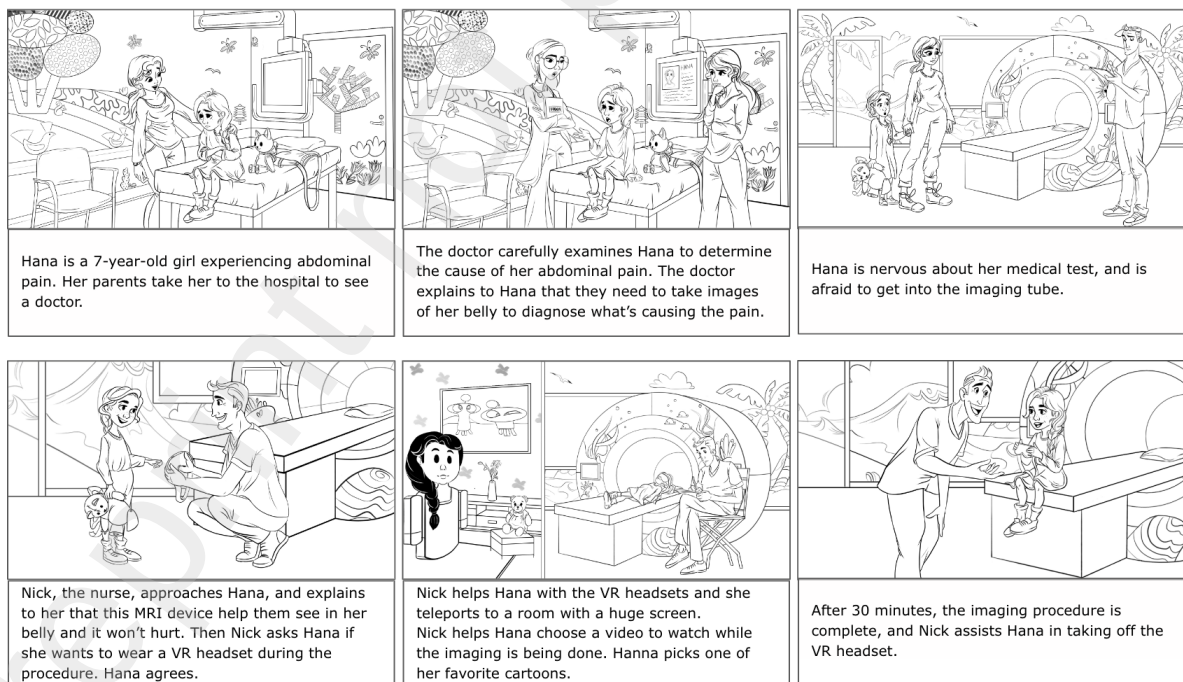


Figure 7: Healthcare Utopia (HU): Use of VR for distraction during children's medical MRI procedures.

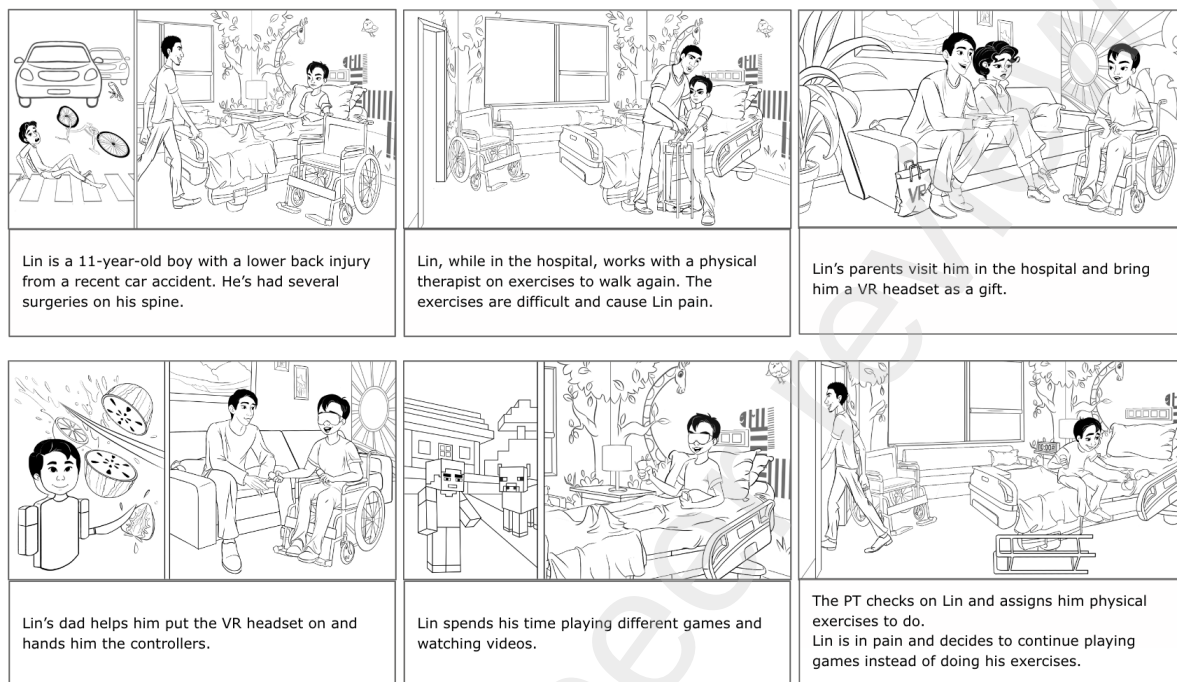


Figure 8: Healthcare Dystopia (HD): Children using VR to cope with pain and avoid physical exertion in post-medical procedures.